

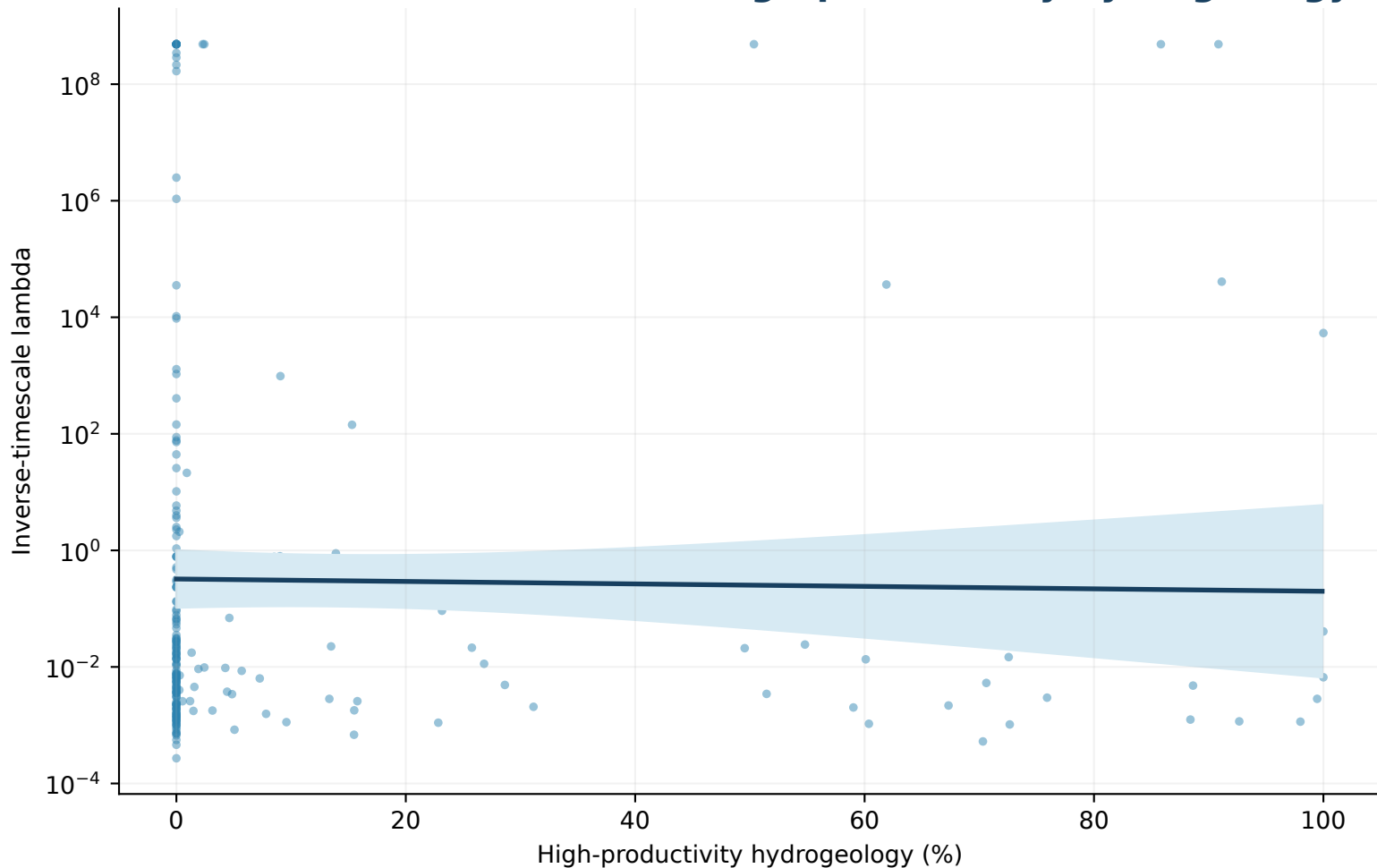
Inverse-timescale lambda one-variable analysis

Target: **inverse characteristic timescale**. Response transformation: $\log(\text{fitted_lambda})$; all eligible values are strictly positive.

Each predictor is fitted separately using OLS with HC3 robust uncertainty. Effects are exact percentage differences for frozen meaningful increments. Raw p-values and Benjamini-Hochberg FDR q-values are both reported.

Primary sample: strict, n range 212-227. Annual NIMROD indices are an independent rainfall FDR family. All associations are unadjusted and non-causal.

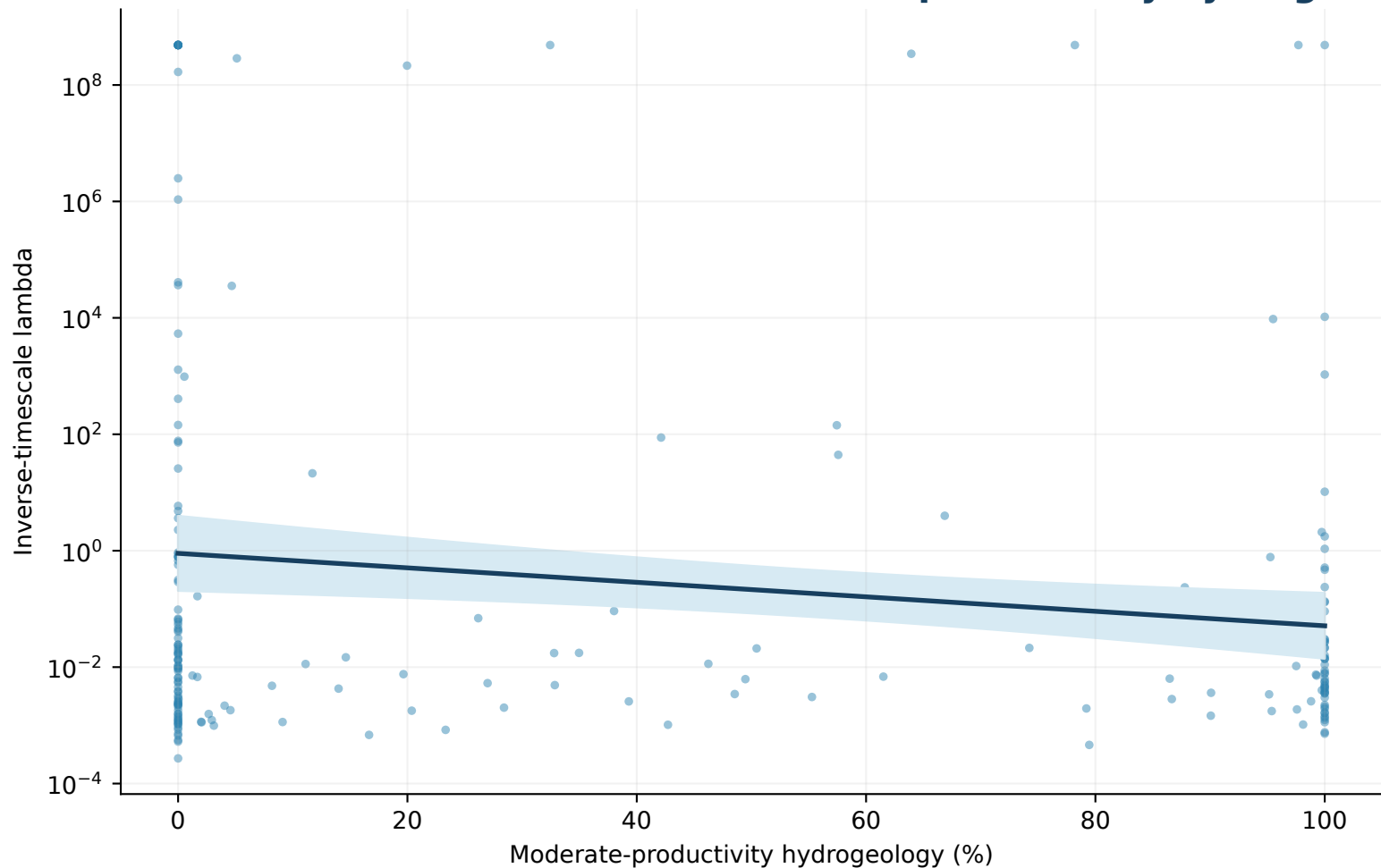
Inverse-timescale lambda vs High-productivity hydrogeology



n=227 | effect -4.7% per 10 percentage points | 95% CI [-34.5, +38.6]% | p=0.8 | q=0.8 | R2=0.000

A 10 percentage points increase in High-productivity hydrogeology is associated with an estimated 4.7% lower inverse-timescale lambda; unadjusted

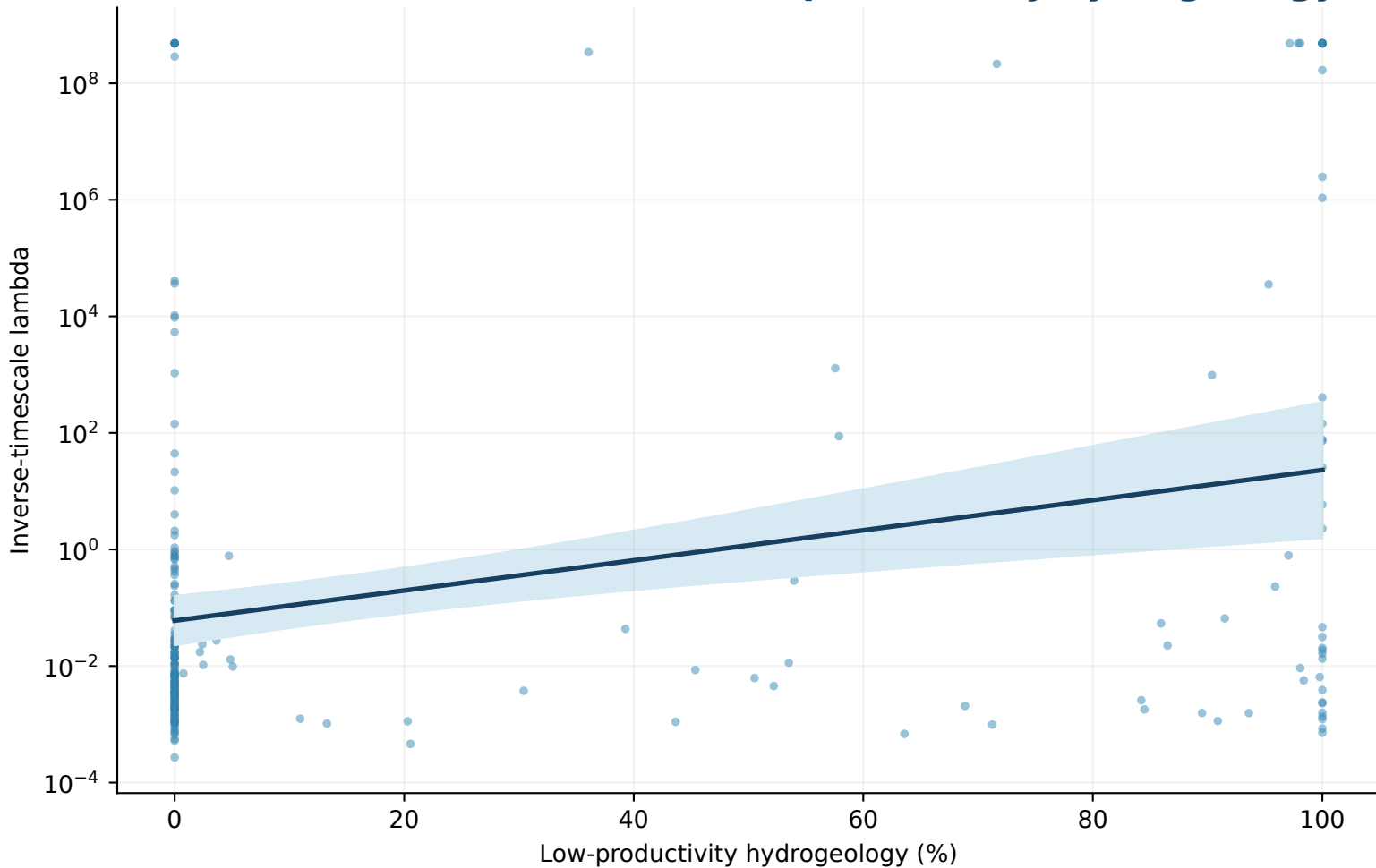
Inverse-timescale lambda vs Moderate-productivity hydrogeology



n=227 | effect -24.9% per 10 percentage points | 95% CI [-38.9, -7.8]% | p=0.00628 | q=0.0117 | R2=0.027

A 10 percentage points increase in Moderate-productivity hydrogeology is associated with an estimated 24.9% lower inverse-timescale lambda; u

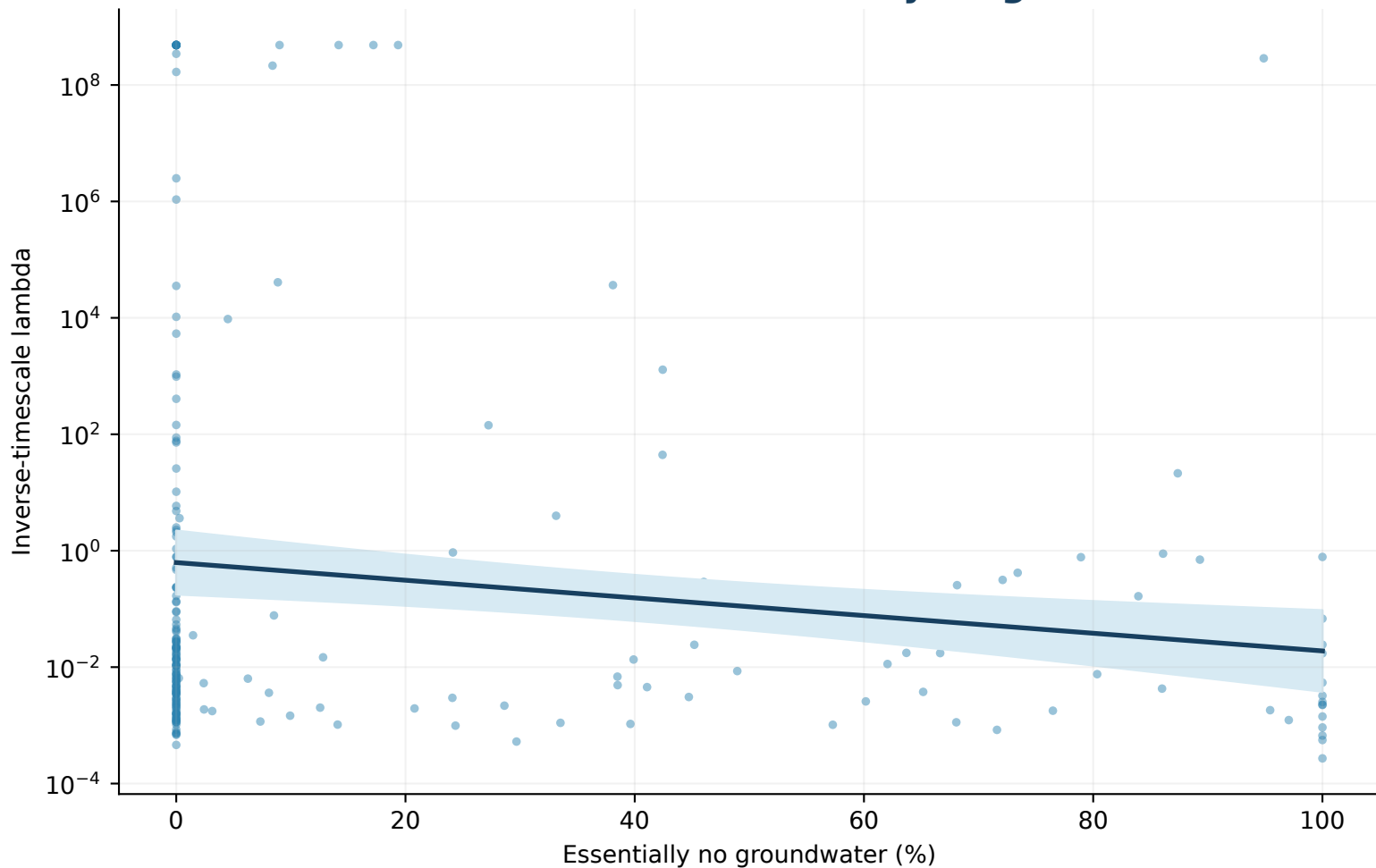
Inverse-timescale lambda vs Low-productivity hydrogeology



n=227 | effect +81.3% per 10 percentage points | 95% CI [+35.1, +143.2]% | p=7.2e-05 | q=0.000356 | R2=0.101

A 10 percentage points increase in Low-productivity hydrogeology is associated with an estimated 81.3% higher inverse-timescale lambda; unadj

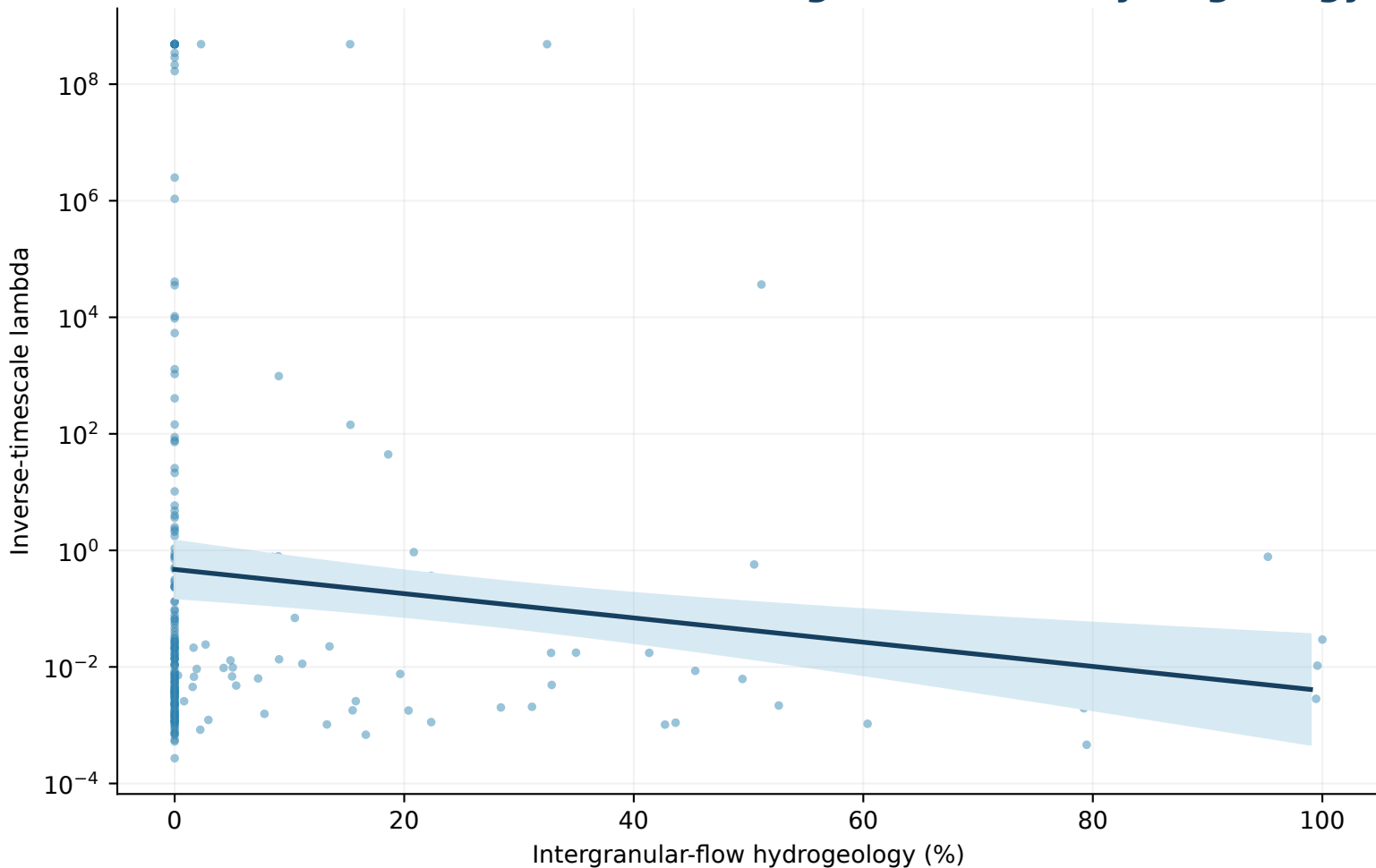
Inverse-timescale lambda vs Essentially no groundwater



n=227 | effect -29.5% per 10 percentage points | 95% CI [-43.4, -12.1]% | p=0.00188 | q=0.00406 | R²=0.024

A 10 percentage points increase in Essentially no groundwater is associated with an estimated 29.5% lower inverse-timescale lambda; unadjusted

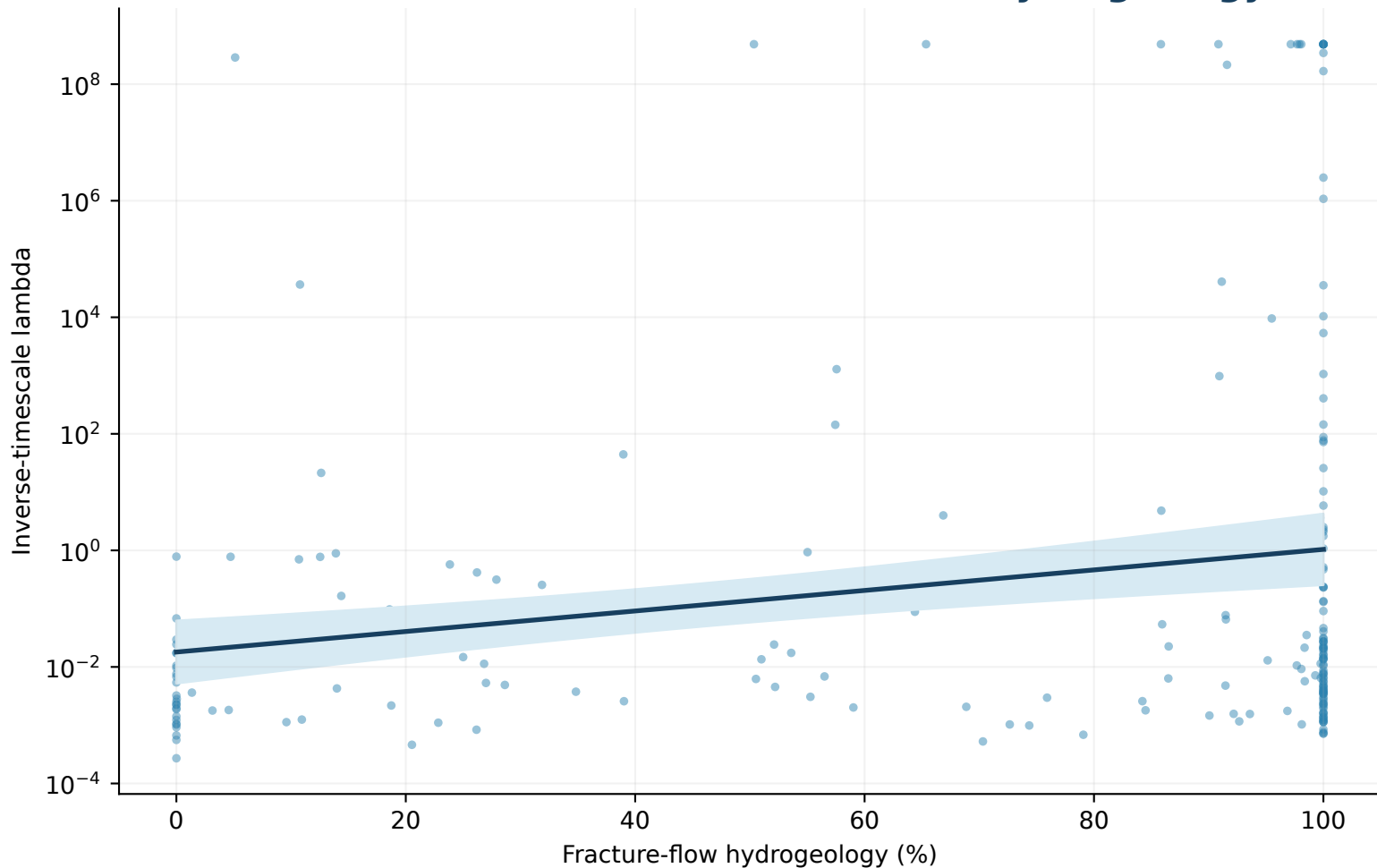
Inverse-timescale lambda vs Intergranular-flow hydrogeology



n=227 | effect -38.1% per 10 percentage points | 95% CI [-52.7, -18.9]% | p=0.000488 | q=0.0014 | R²=0.018

A 10 percentage points increase in Intergranular-flow hydrogeology is associated with an estimated 38.1% lower inverse-timescale lambda; unadjusted

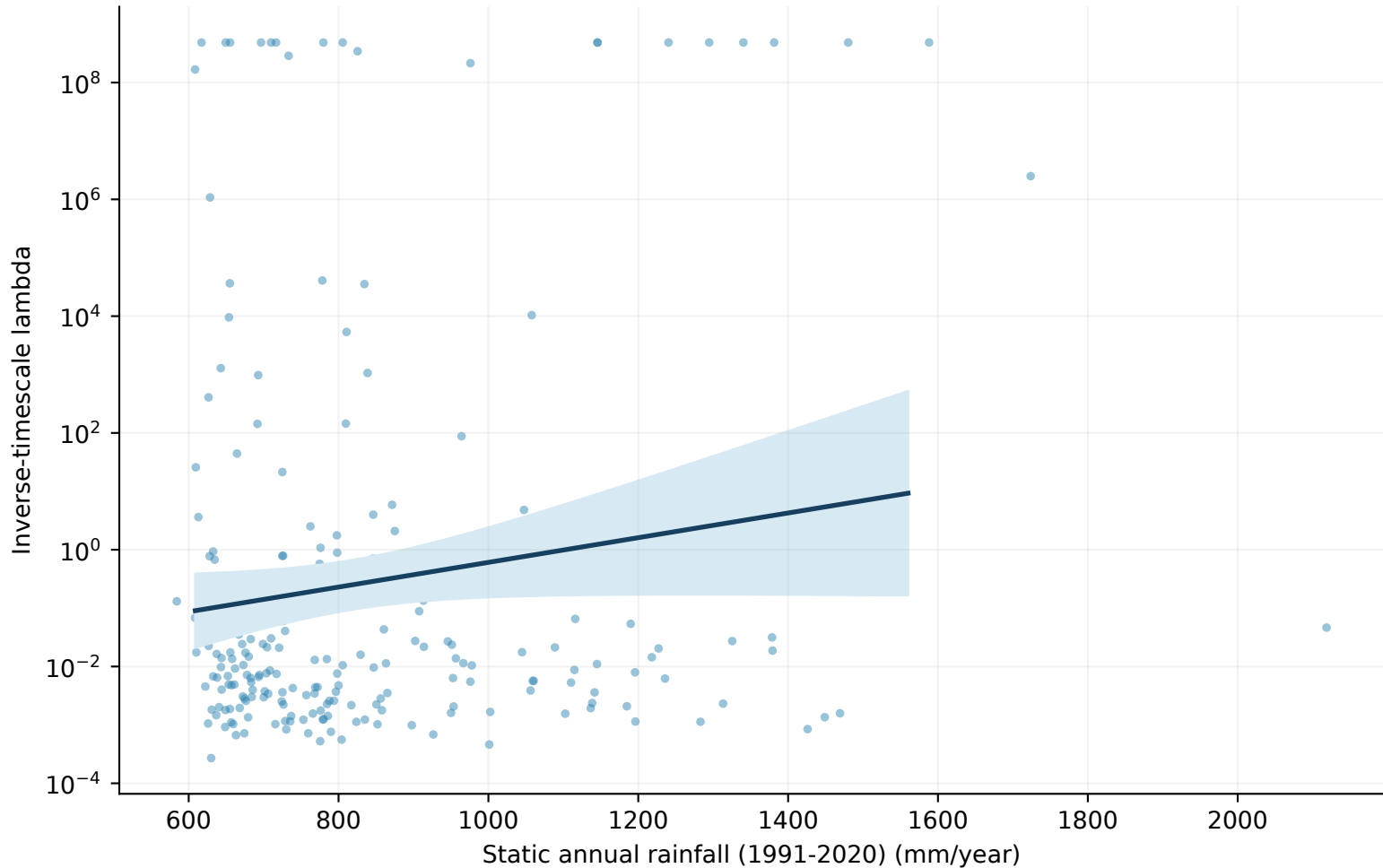
Inverse-timescale lambda vs Fracture-flow hydrogeology



n=227 | effect +50.0% per 10 percentage points | 95% CI [+22.6, +83.6]% | p=8.21e-05 | q=0.000356 | R2=0.043

A 10 percentage points increase in Fracture-flow hydrogeology is associated with an estimated 50.0% higher inverse-timescale lambda; unadjusted

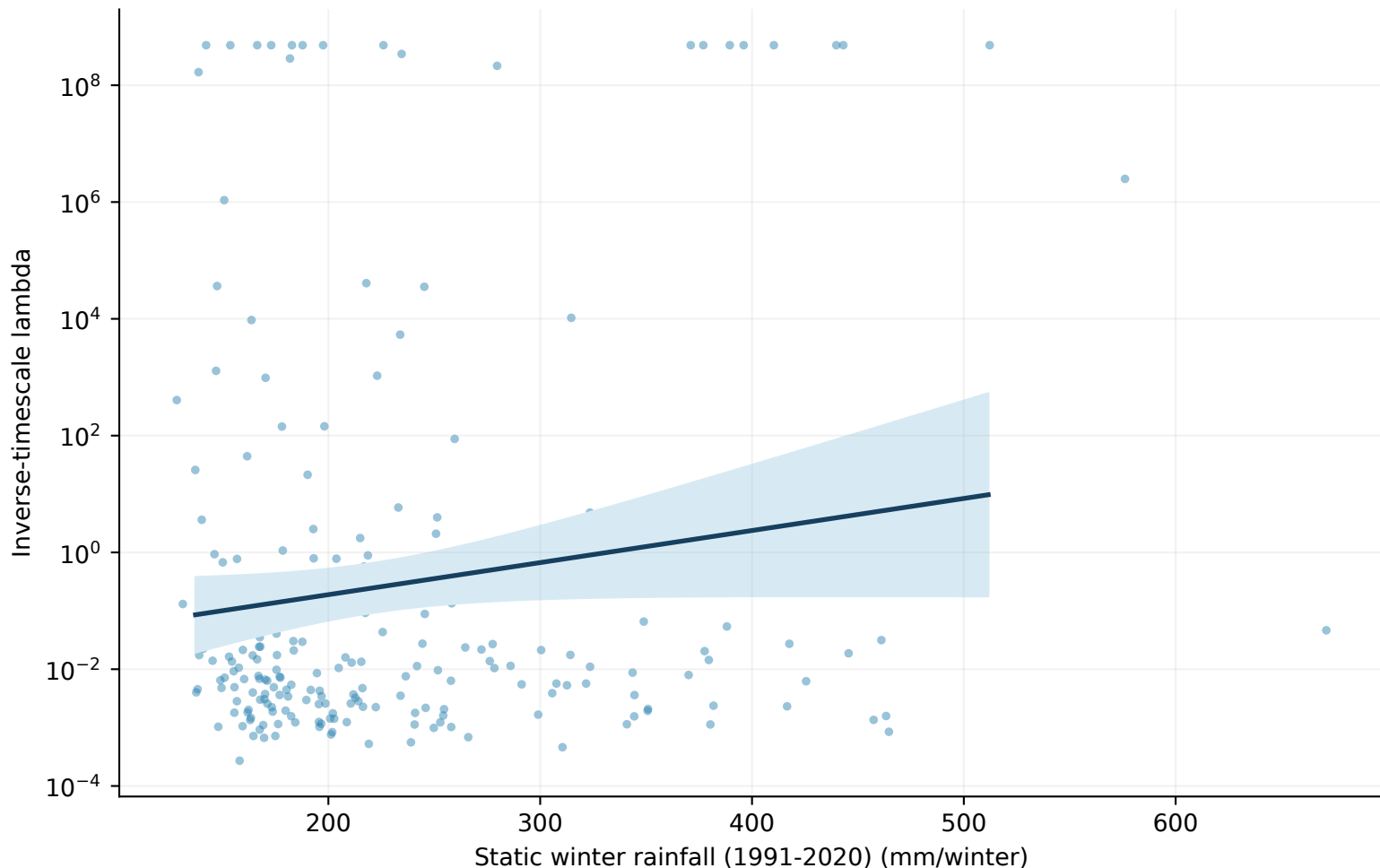
Inverse-timescale lambda vs Static annual rainfall (1991-2020)



n=227 | effect +62.7% per 100 mm/year | 95% CI [-3.7, +175.0]% | p=0.069 | q=0.0996 | R2=0.024

A 100 mm/year increase in Static annual rainfall (1991-2020) is associated with an estimated 62.7% higher inverse-timescale lambda; unadjusted

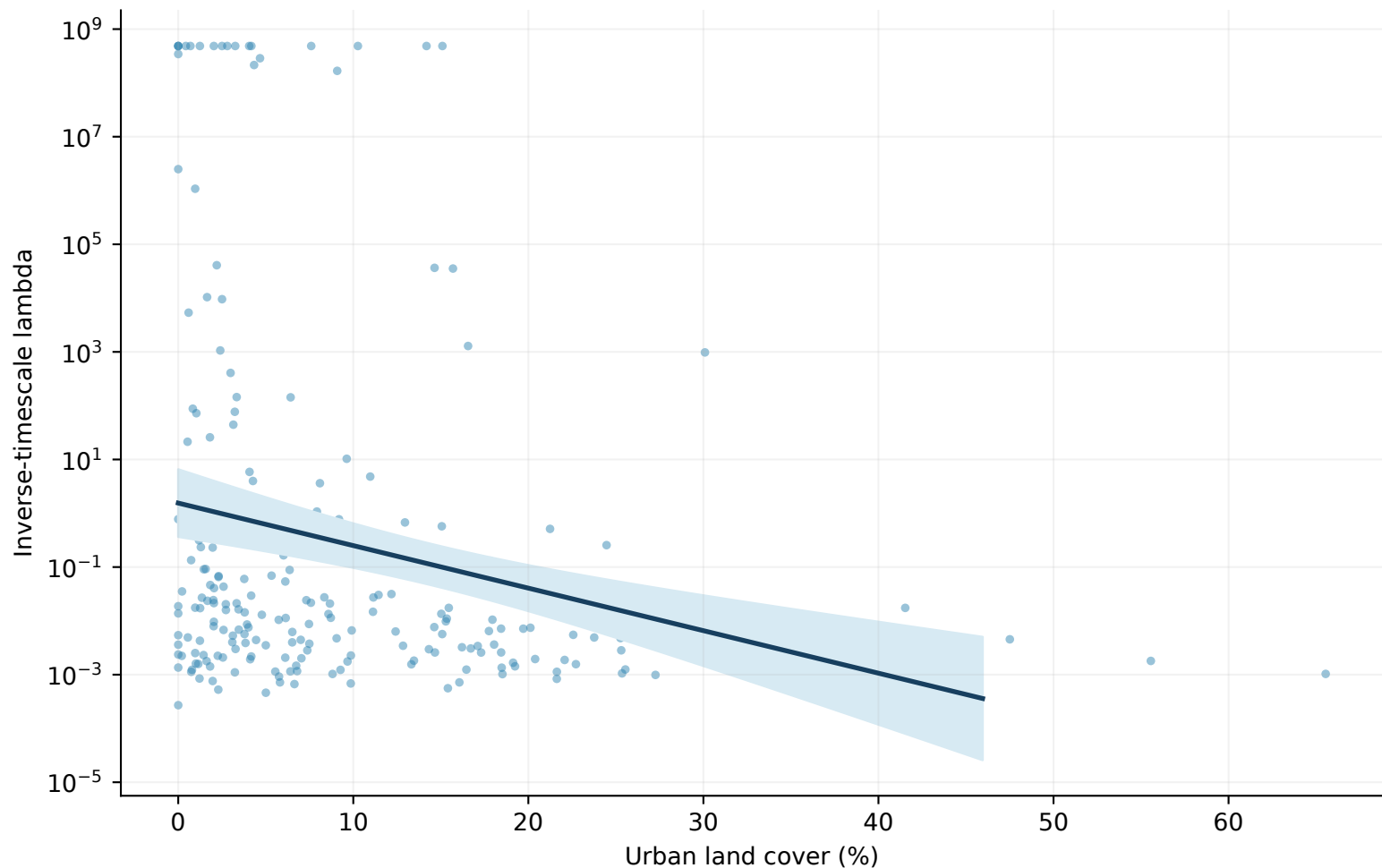
Inverse-timescale lambda vs Static winter rainfall (1991-2020)



n=227 | effect +254.2% per 100 mm/winter | 95% CI [-6.8, +1245.3]% | p=0.0633 | q=0.0996 | R2=0.024

A 100 mm/winter increase in Static winter rainfall (1991-2020) is associated with an estimated 254.2% higher inverse-timescale lambda; unadjusted

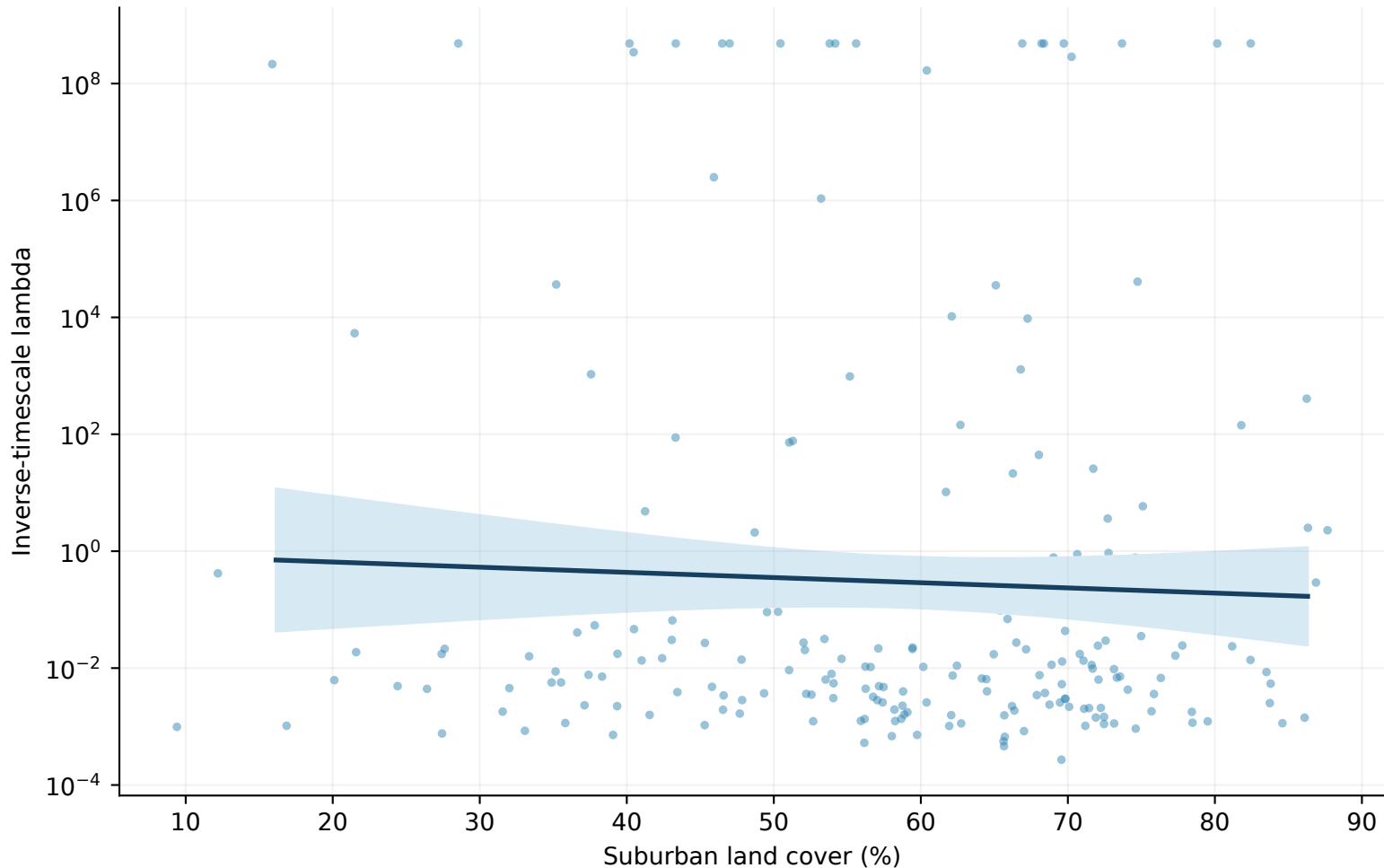
Inverse-timescale lambda vs Urban land cover



n=227 | effect -83.8% per 10 percentage points | 95% CI [-92.6, -64.3]% | p=6.21e-06 | q=8.07e-05 | R2=0.053

A 10 percentage points increase in Urban land cover is associated with an estimated 83.8% lower inverse-timescale lambda; unadjusted and non

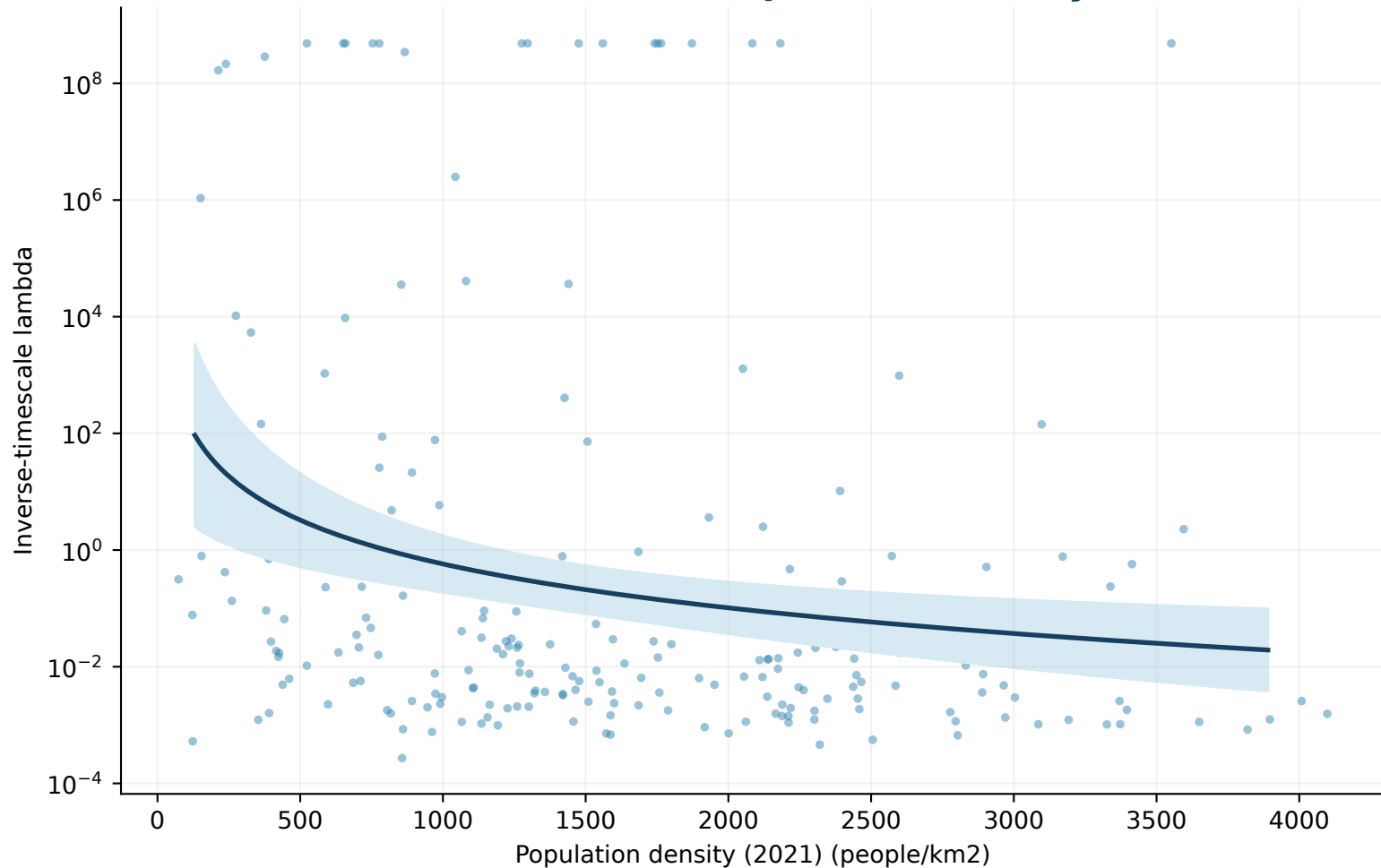
Inverse-timescale lambda vs Suburban land cover



n=227 | effect -18.5% per 10 percentage points | 95% CI [-55.6, +49.9]% | p=0.511 | q=0.554 | R2=0.002

A 10 percentage points increase in Suburban land cover is associated with an estimated 18.5% lower inverse-timescale lambda; unadjusted and

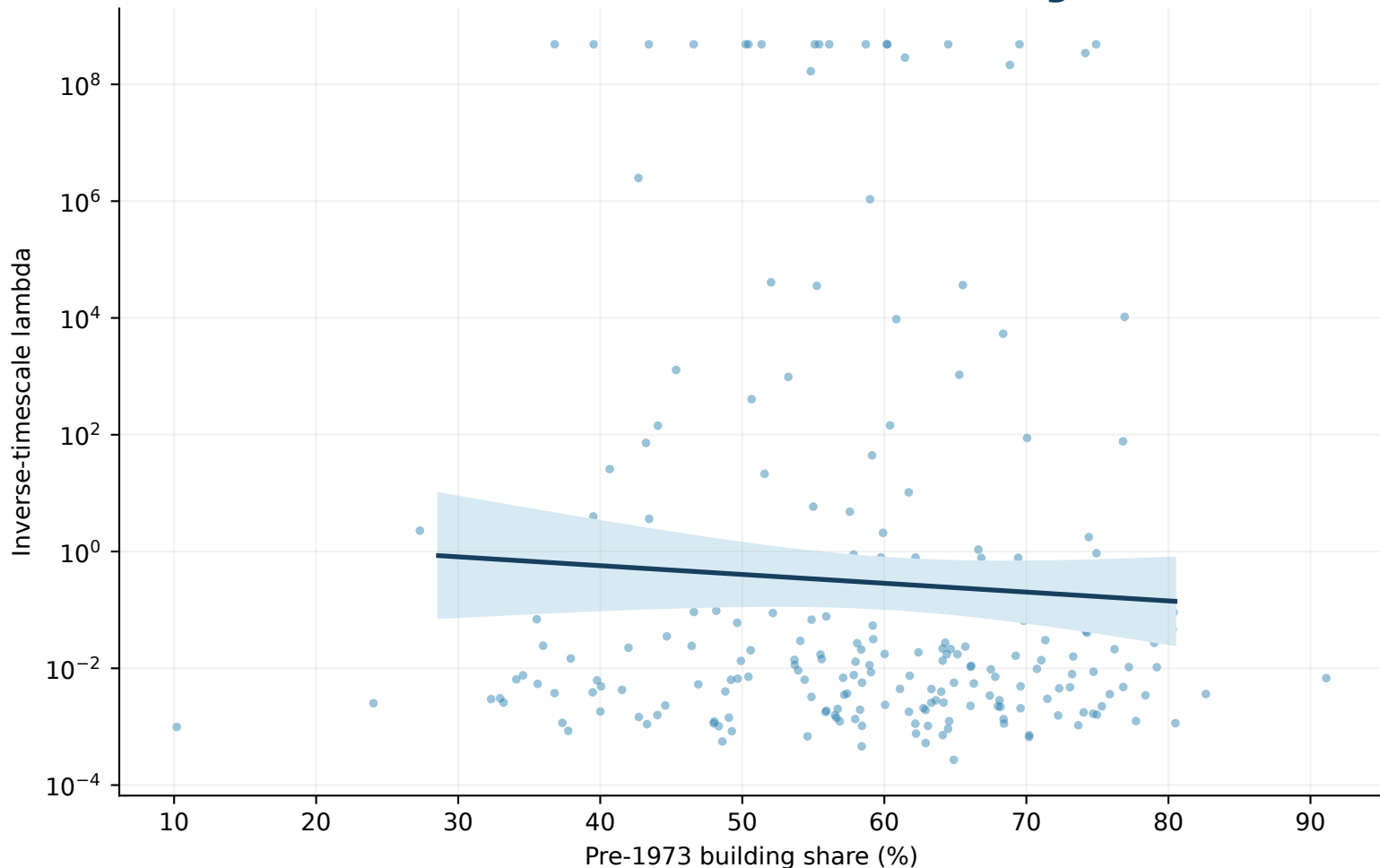
Inverse-timescale lambda vs Population density (2021)



n=227 | effect -82.3% per doubling | 95% CI [-93.4, -52.8]% | p=0.00054 | q=0.0014 | R²=0.058

A doubling increase in Population density (2021) is associated with an estimated 82.3% lower inverse-timescale lambda; unadjusted and non-cau

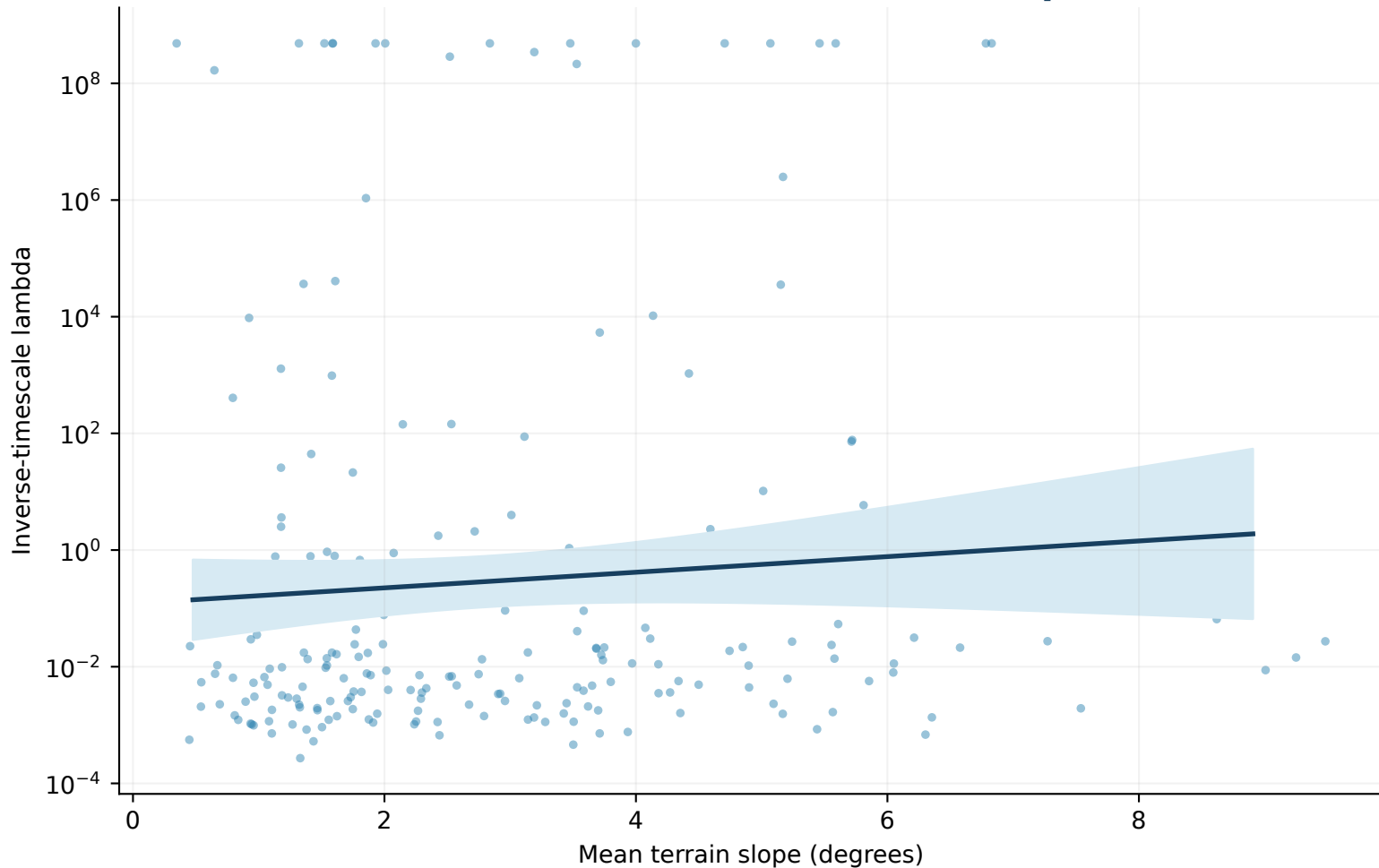
Inverse-timescale lambda vs Pre-1973 building share



n=227 | effect -29.4% per 10 percentage points | 95% CI [-65.0, +42.7]% | p=0.333 | q=0.393 | R2=0.003

A 10 percentage points increase in Pre-1973 building share is associated with an estimated 29.4% lower inverse-timescale lambda; unadjusted and

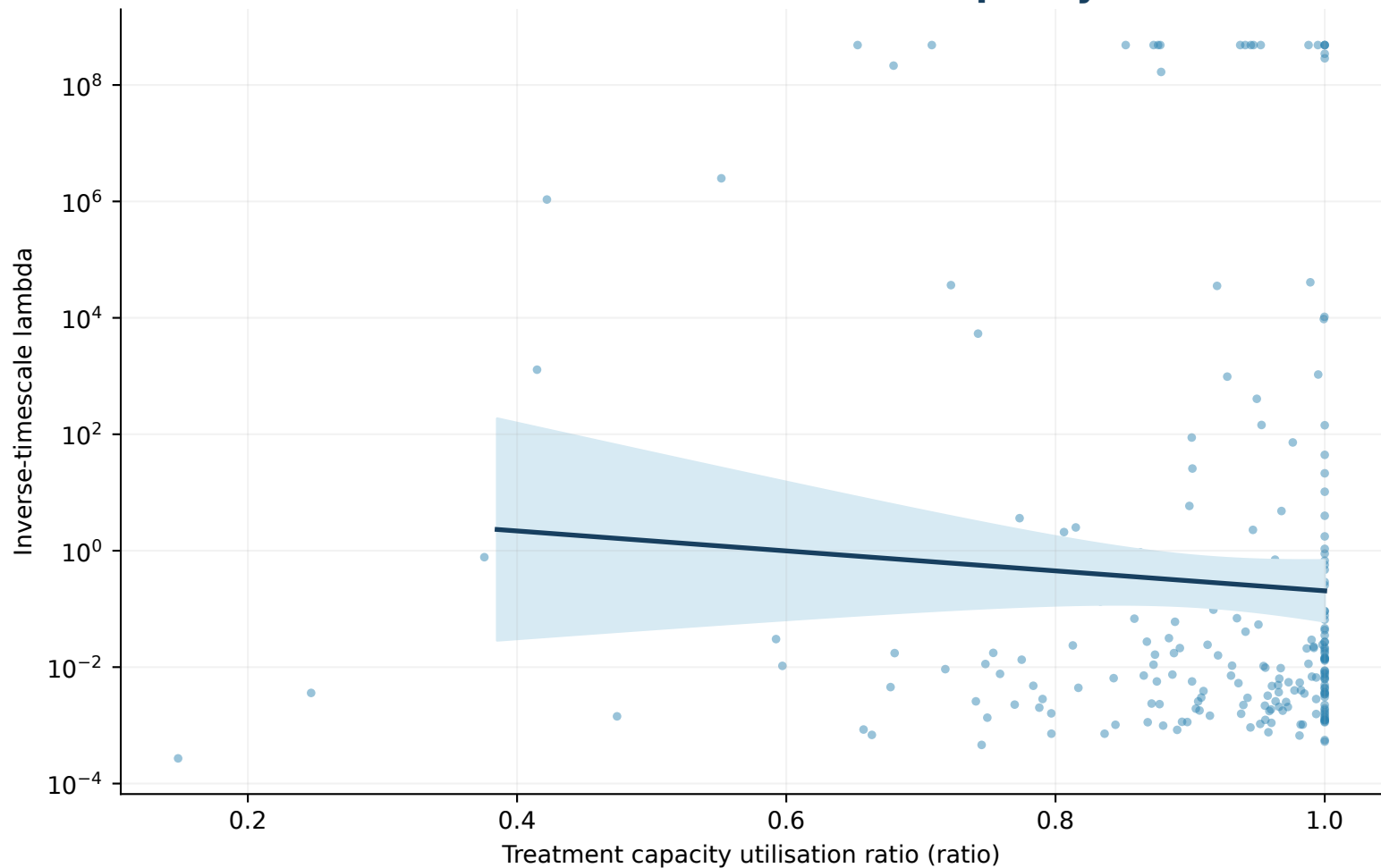
Inverse-timescale lambda vs Mean terrain slope



n=227 | effect +36.1% per 1 degree | 95% CI [-19.4, +130.0]% | p=0.249 | q=0.324 | R2=0.006

A 1 degree increase in Mean terrain slope is associated with an estimated 36.1% higher inverse-timescale lambda; unadjusted and non-causal.

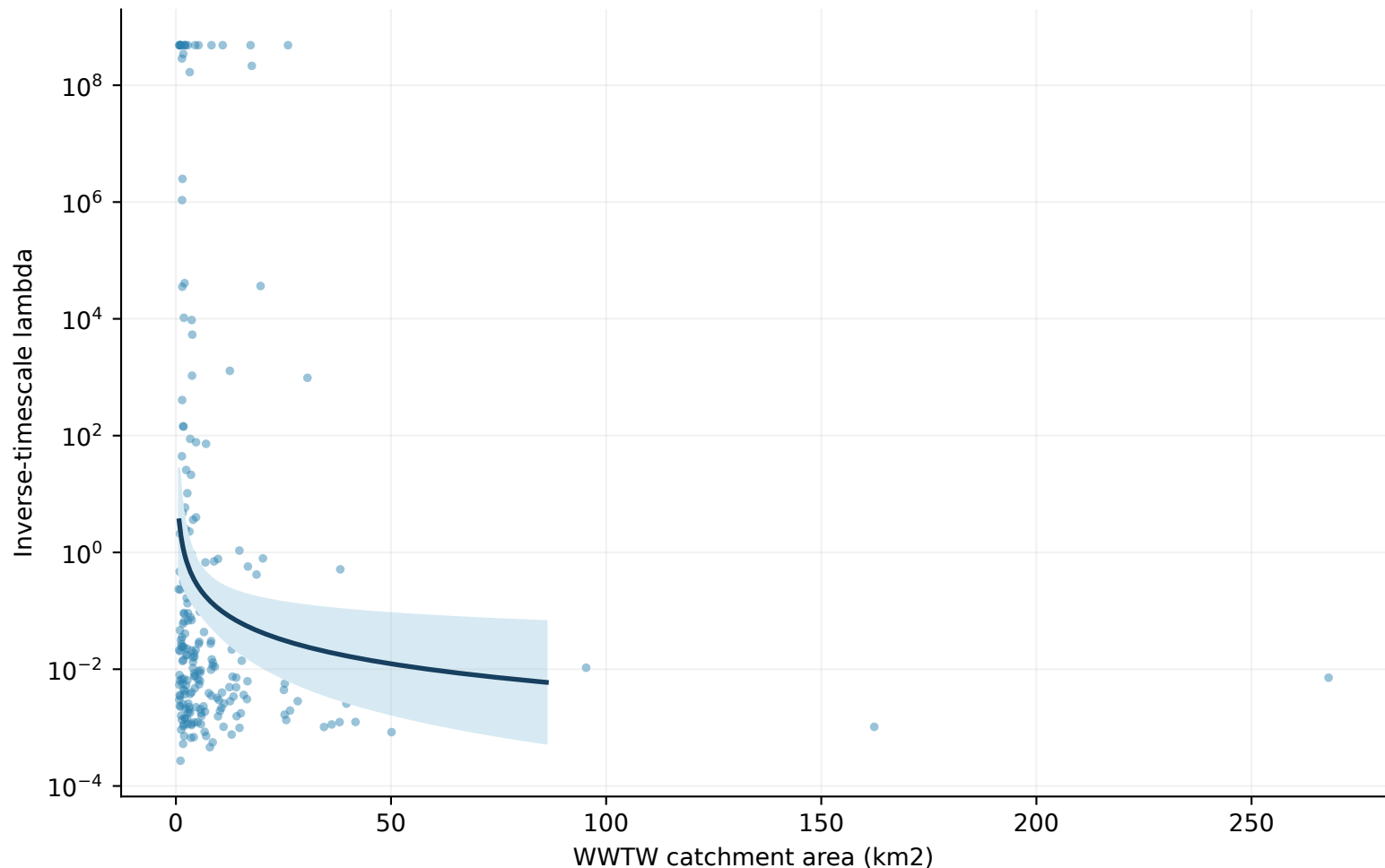
Inverse-timescale lambda vs Treatment capacity utilisation ratio



n=227 | effect -32.6% per 0.1 ratio units | 95% CI [-69.9, +50.8]% | p=0.337 | q=0.337 | R2=0.005

A 0.1 ratio units increase in Treatment capacity utilisation ratio is associated with an estimated 32.6% lower inverse-timescale lambda; unadjusted

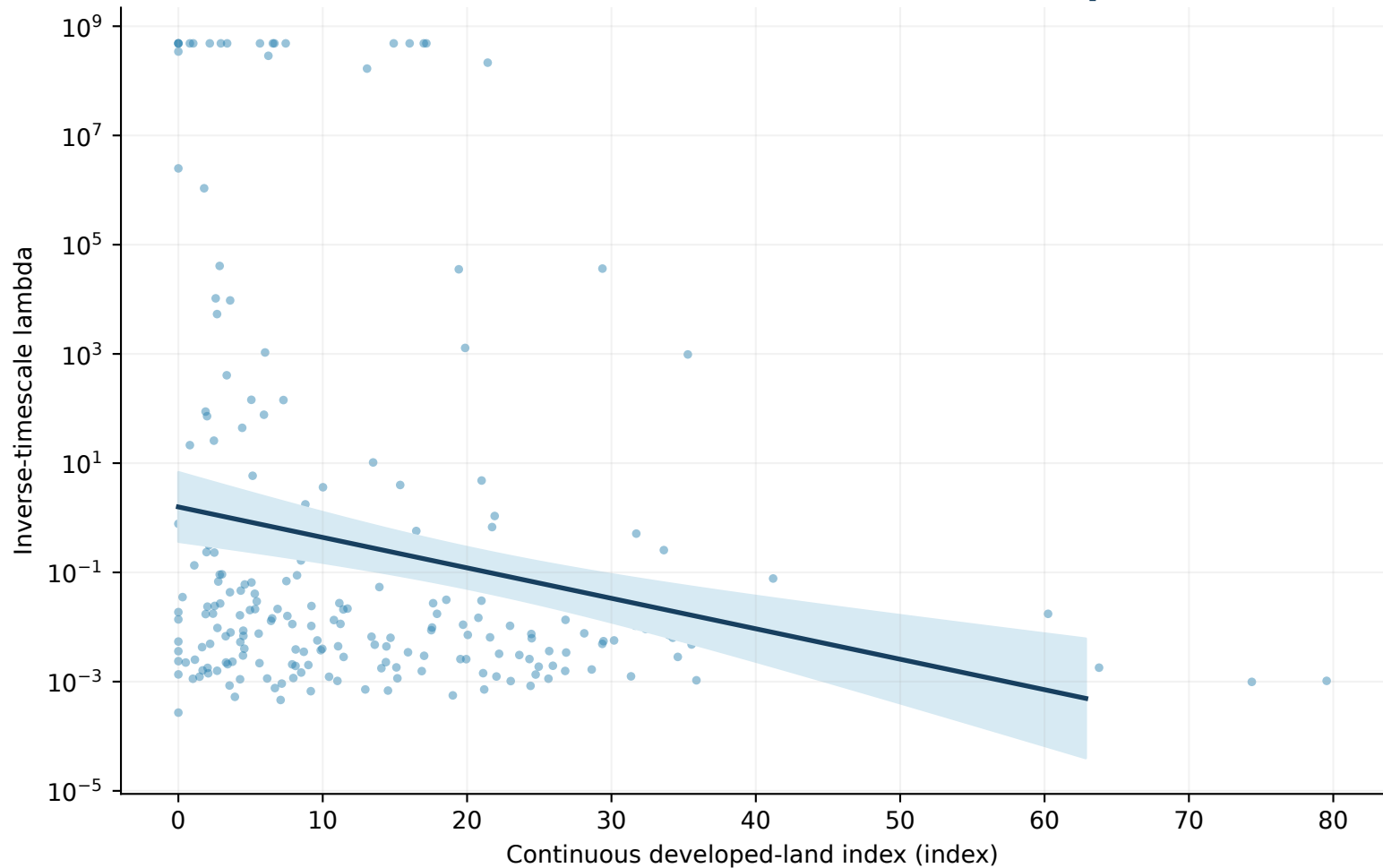
Inverse-timescale lambda vs WWTW catchment area



n=227 | effect -60.6% per doubling | 95% CI [-78.2, -28.8]% | p=0.00204 | q=0.00407 | R²=0.040

A doubling increase in WWTW catchment area is associated with an estimated 60.6% lower inverse-timescale lambda; unadjusted and non-causal

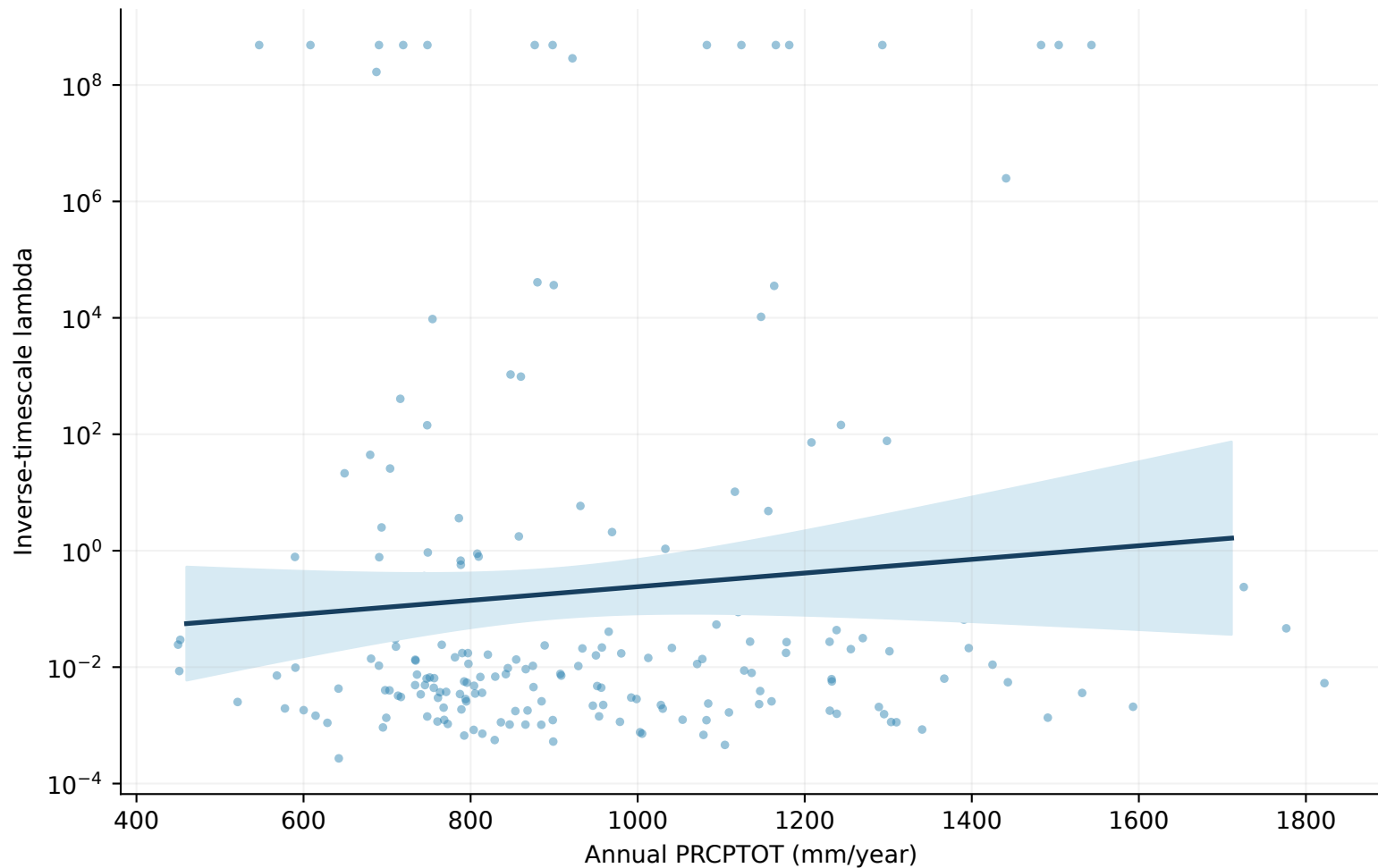
Inverse-timescale lambda vs Continuous developed-land index



n=227 | effect -72.3% per 10 index points | 95% CI [-84.2, -51.4]% | p=7.83e-06 | q=7.83e-06 | R2=0.047

A 10 index points increase in Continuous developed-land index is associated with an estimated 72.3% lower inverse-timescale lambda; unadjusted

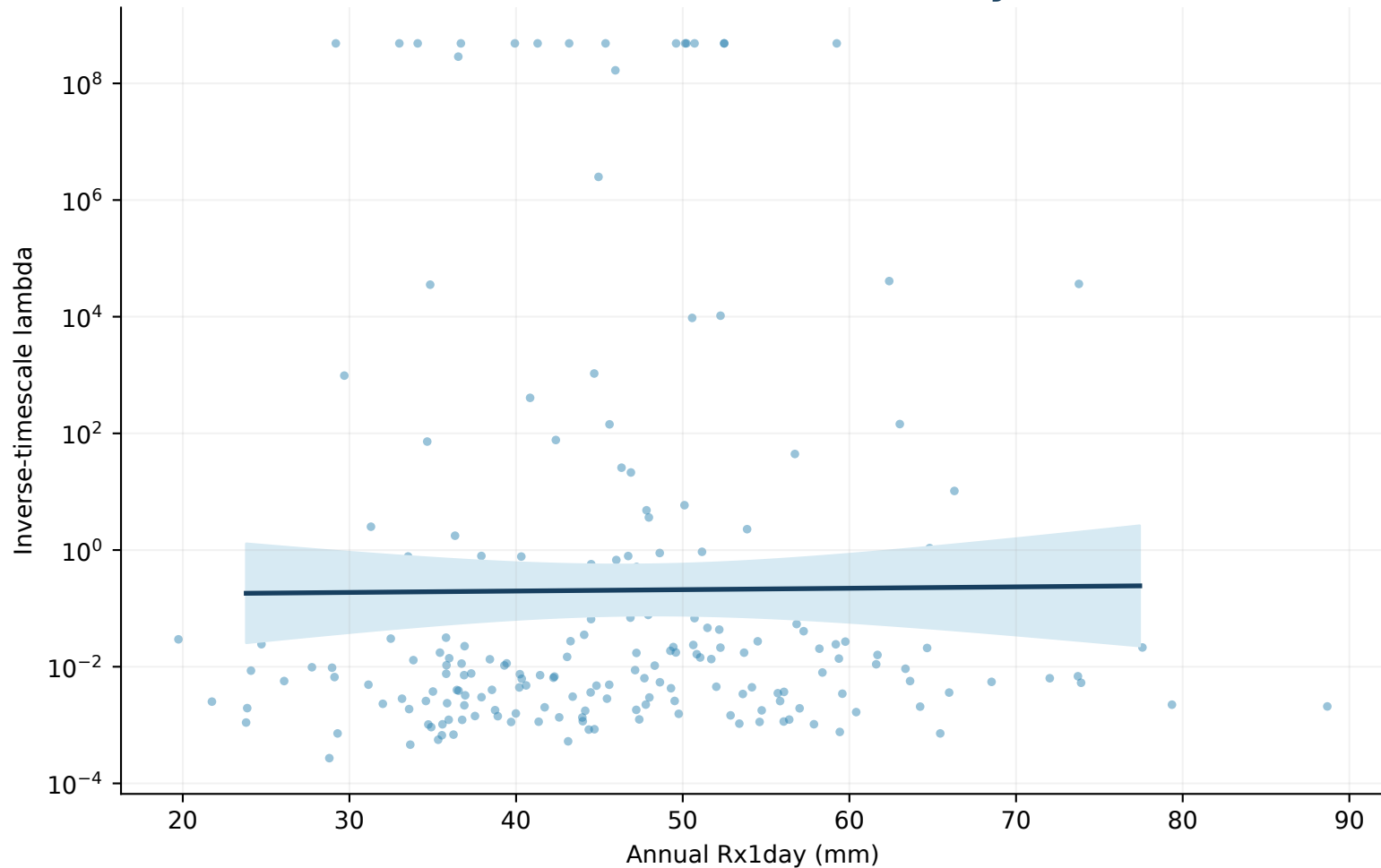
Inverse-timescale lambda vs Annual PRCPTOT



n=212 | effect +31.0% per 100 mm/year | 95% CI [-16.9, +106.6]% | p=0.245 | q=0.392 | R2=0.009

A 100 mm/year increase in Annual PRCPTOT is associated with an estimated 31.0% higher inverse-timescale lambda; unadjusted and non-causal.

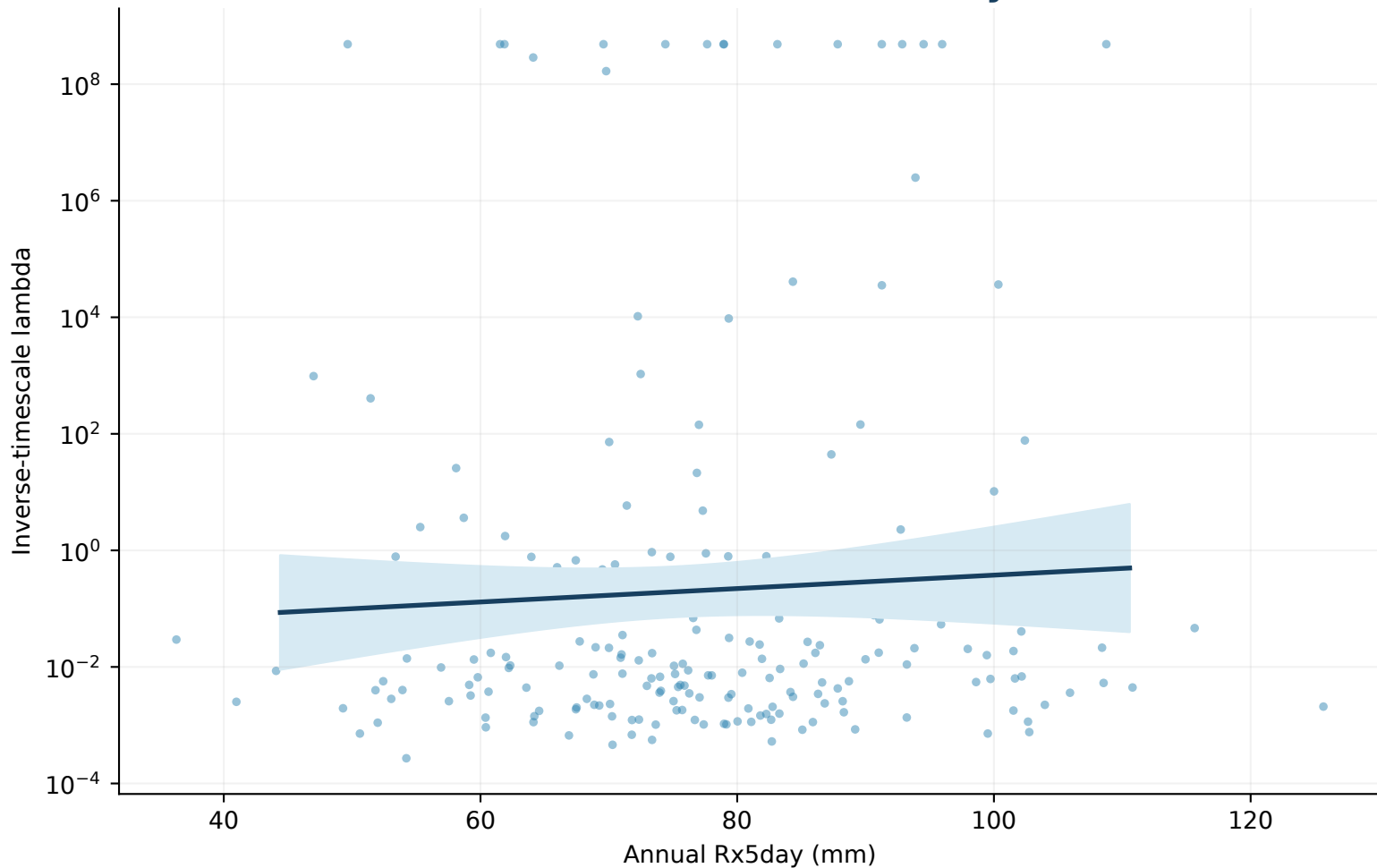
Inverse-timescale lambda vs Annual Rx1day



n=212 | effect +5.6% per 10 mm | 95% CI [-48.4, +116.4]% | p=0.881 | q=0.881 | R2=0.000

A 10 mm increase in Annual Rx1day is associated with an estimated 5.6% higher inverse-timescale lambda; unadjusted and non-causal.

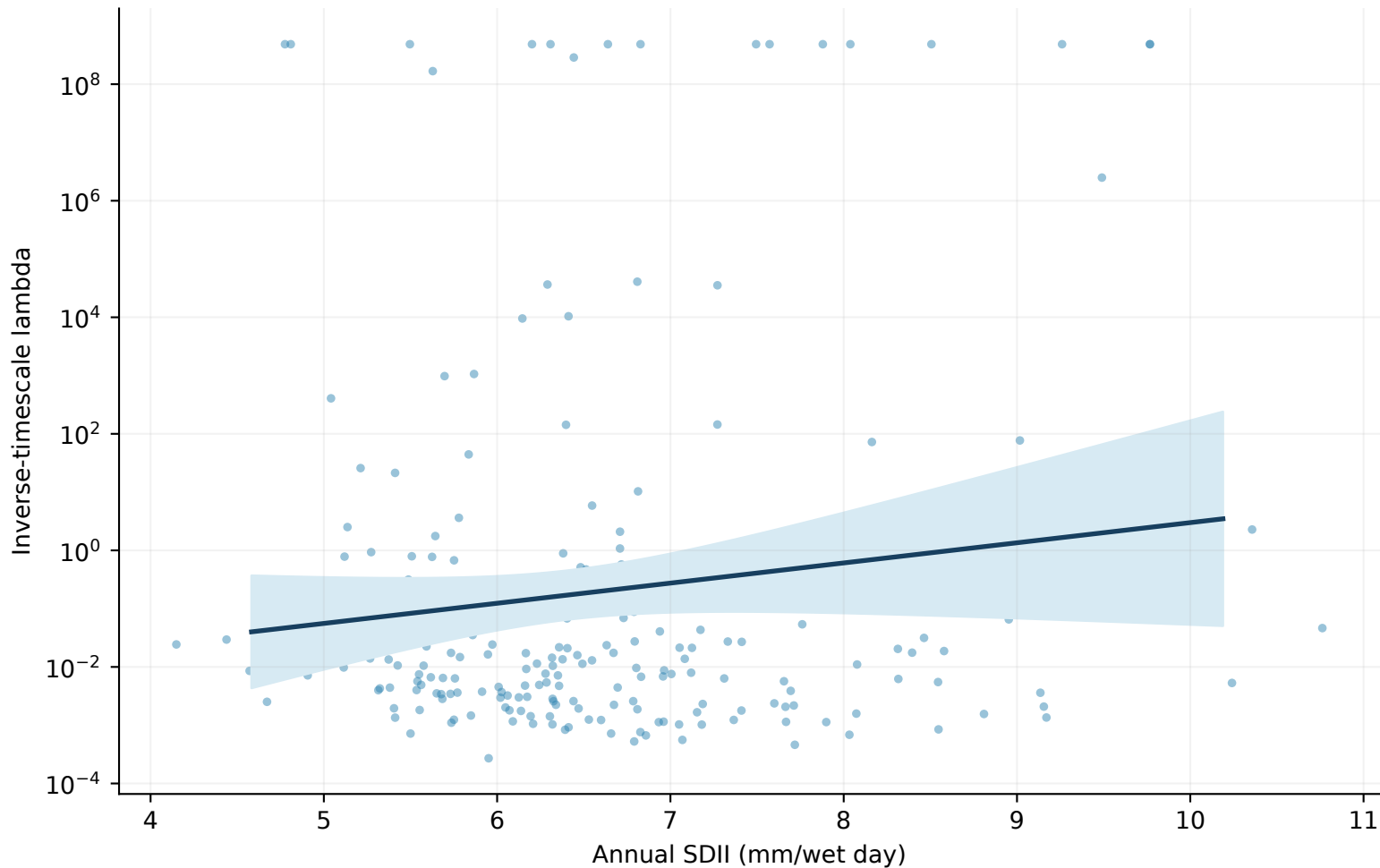
Inverse-timescale lambda vs Annual Rx5day



n=212 | effect +94.1% per 25 mm | 95% CI [-62.4, +902.5]% | p=0.429 | q=0.49 | R2=0.003

A 25 mm increase in Annual Rx5day is associated with an estimated 94.1% higher inverse-timescale lambda; unadjusted and non-causal.

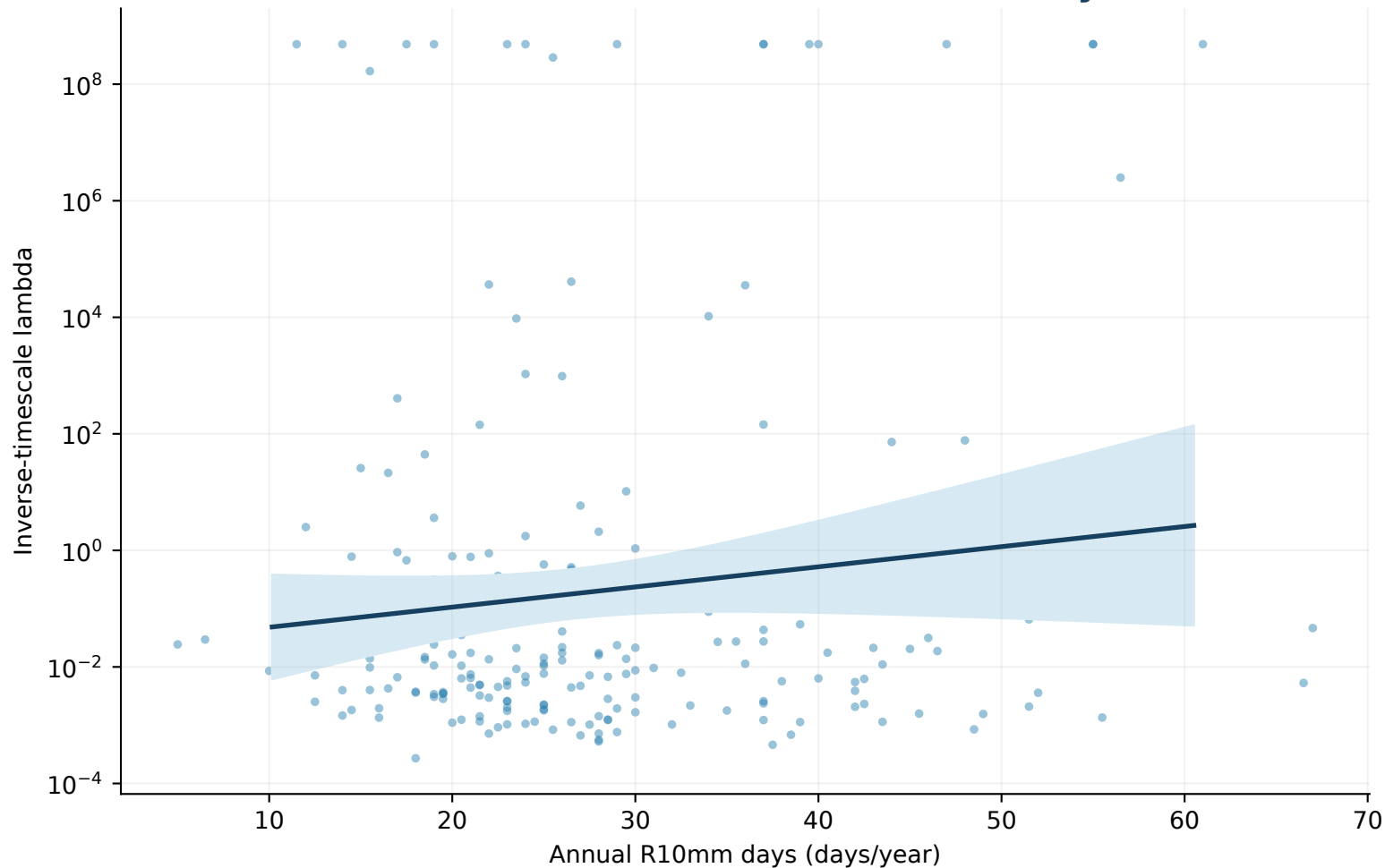
Inverse-timescale lambda vs Annual SDII



n=212 | effect +121.8% per 1 mm/wet day | 95% CI [-25.5, +560.5]% | p=0.152 | q=0.327 | R2=0.016

A 1 mm/wet day increase in Annual SDII is associated with an estimated 121.8% higher inverse-timescale lambda; unadjusted and non-causal.

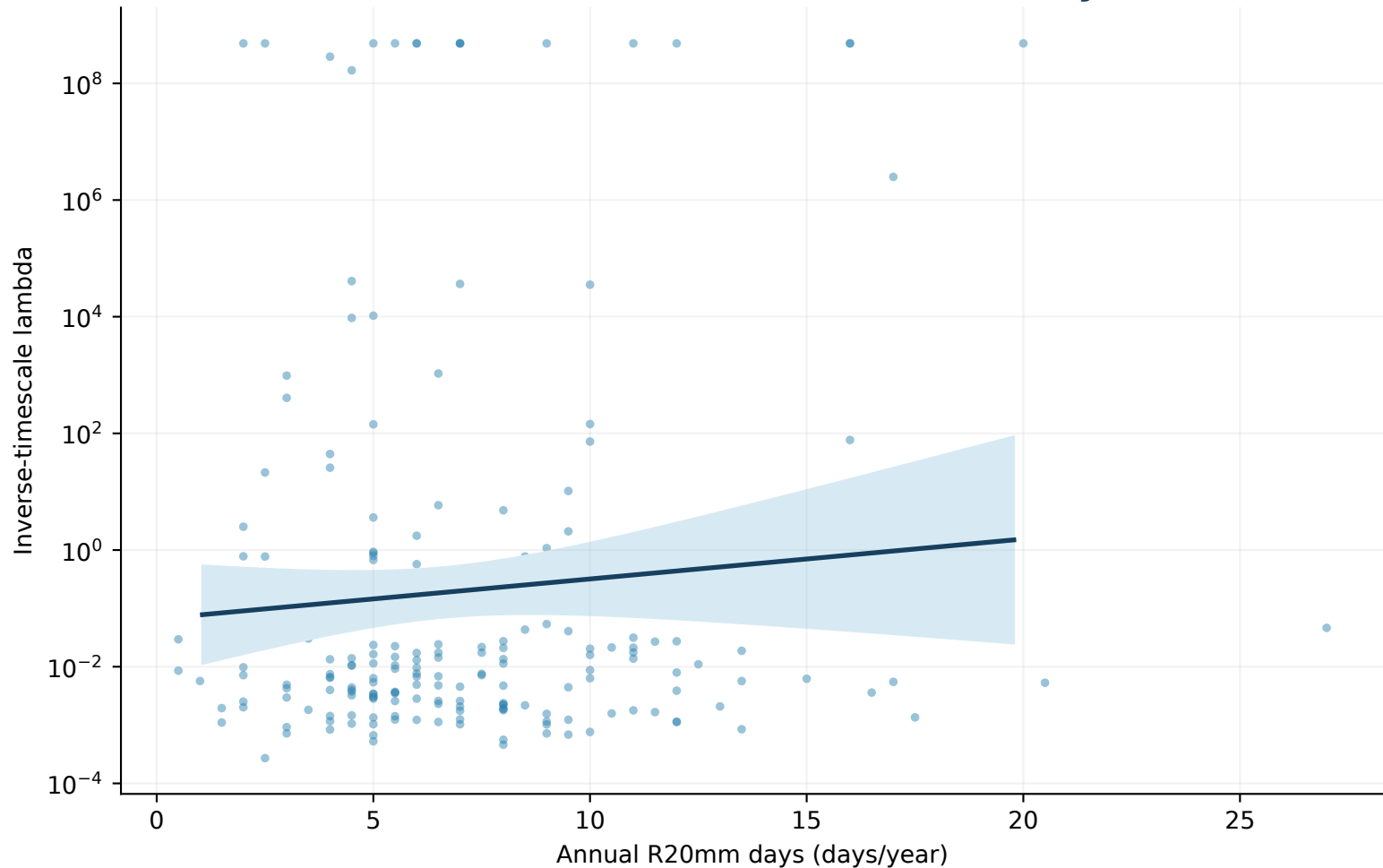
Inverse-timescale lambda vs Annual R10mm days



n=212 | effect +48.9% per 5 days/year | 95% CI [-15.0, +160.8]% | p=0.164 | q=0.327 | R2=0.015

A 5 days/year increase in Annual R10mm days is associated with an estimated 48.9% higher inverse-timescale lambda; unadjusted and non-causal

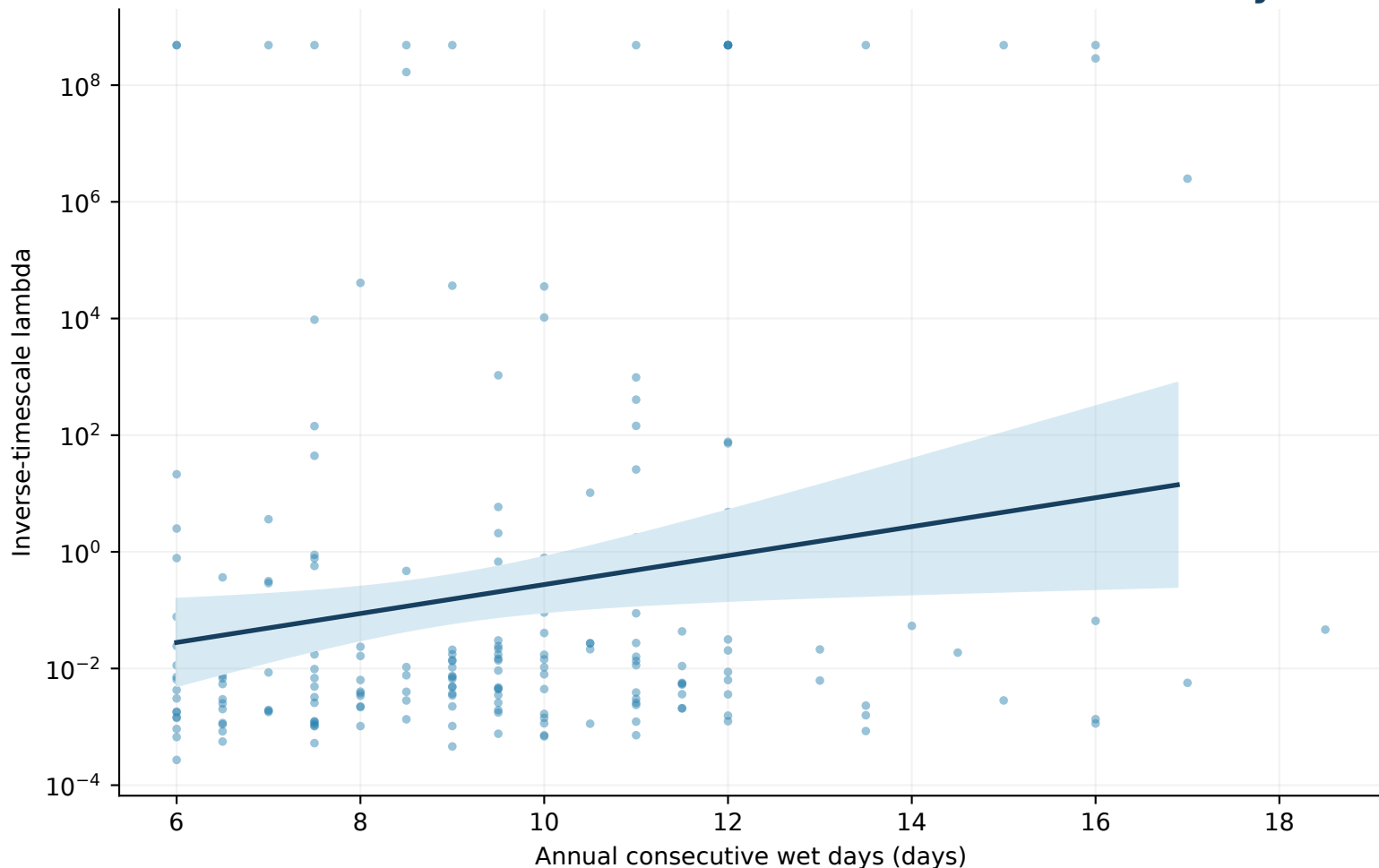
Inverse-timescale lambda vs Annual R20mm days



n=212 | effect +119.9% per 5 days/year | 95% CI [-51.0, +886.5]% | p=0.303 | q=0.405 | R²=0.007

A 5 days/year increase in Annual R20mm days is associated with an estimated 119.9% higher inverse-timescale lambda; unadjusted and non-causal

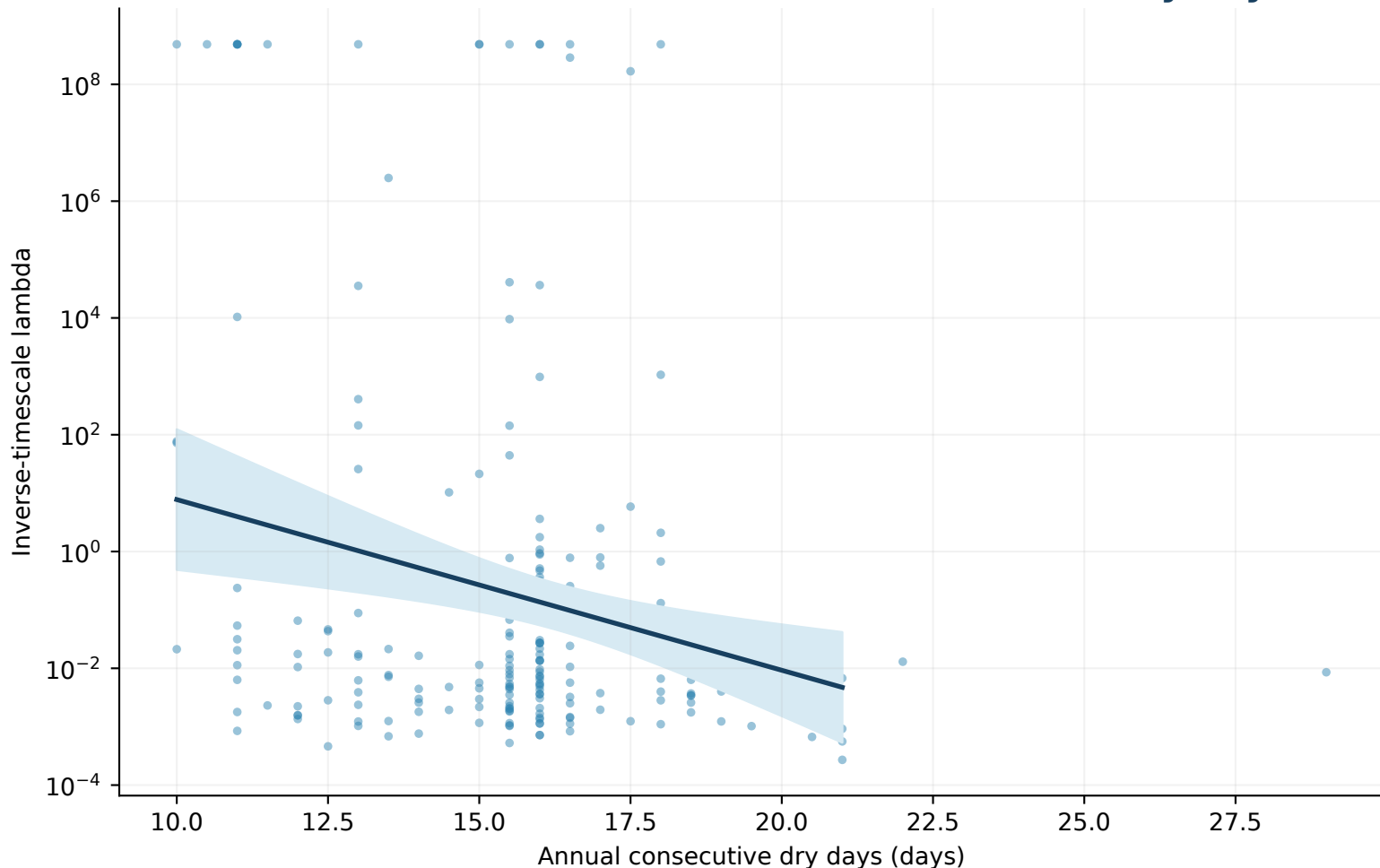
Inverse-timescale lambda vs Annual consecutive wet days



n=212 | effect +77.1% per 1 day | 95% CI [+8.4, +189.1]% | p=0.0224 | q=0.0895 | R2=0.039

A 1 day increase in Annual consecutive wet days is associated with an estimated 77.1% higher inverse-timescale lambda; unadjusted and non-ca

Inverse-timescale lambda vs Annual consecutive dry days



n=212 | effect -96.5% per 5 days | 95% CI [-99.6, -72.2]% | p=0.00156 | q=0.0125 | R2=0.056

A 5 days increase in Annual consecutive dry days is associated with an estimated 96.5% lower inverse-timescale lambda; unadjusted and non-causal

Results summary

Continuous developed-land index: -72.3% [-84.2, -51.4]%; $p=7.83e-06$; $q=7.83e-06$; FDR-supported.

Urban land cover: -83.8% [-92.6, -64.3]%; $p=6.21e-06$; $q=8.07e-05$; FDR-supported.

Low-productivity hydrogeology: +81.3% [+35.1, +143.2]%; $p=7.2e-05$; $q=0.000356$; FDR-supported.

Fracture-flow hydrogeology: +50.0% [+22.6, +83.6]%; $p=8.21e-05$; $q=0.000356$; FDR-supported.

Intergranular-flow hydrogeology: -38.1% [-52.7, -18.9]%; $p=0.000488$; $q=0.0014$; FDR-supported.

Population density (2021): -82.3% [-93.4, -52.8]%; $p=0.00054$; $q=0.0014$; FDR-supported.

Essentially no groundwater: -29.5% [-43.4, -12.1]%; $p=0.00188$; $q=0.00406$; FDR-supported.

WWTW catchment area: -60.6% [-78.2, -28.8]%; $p=0.00204$; $q=0.00407$; FDR-supported.

Moderate-productivity hydrogeology: -24.9% [-38.9, -7.8]%; $p=0.00628$; $q=0.0117$; FDR-supported.

Annual consecutive dry days: -96.5% [-99.6, -72.2]%; $p=0.00156$; $q=0.0125$; FDR-supported.

Annual consecutive wet days: +77.1% [+8.4, +189.1]%; $p=0.0224$; $q=0.0895$; nominal only.

Static annual rainfall (1991-2020): +62.7% [-3.7, +175.0]%; $p=0.069$; $q=0.0996$; no clear evidence.